FINC 4101
Investment Analysis

Topic: Term Structure Analysis

Learning Objectives

1. Define term structure of interest rates and identify common yield-curve shapes.
2. Explain how expected future short-term rates affect the yield curve under Expectations Theory.
3. Calculate and interpret one-year forward rates.
4. Explain how a liquidity premium affects the interpretation of the yield curve.

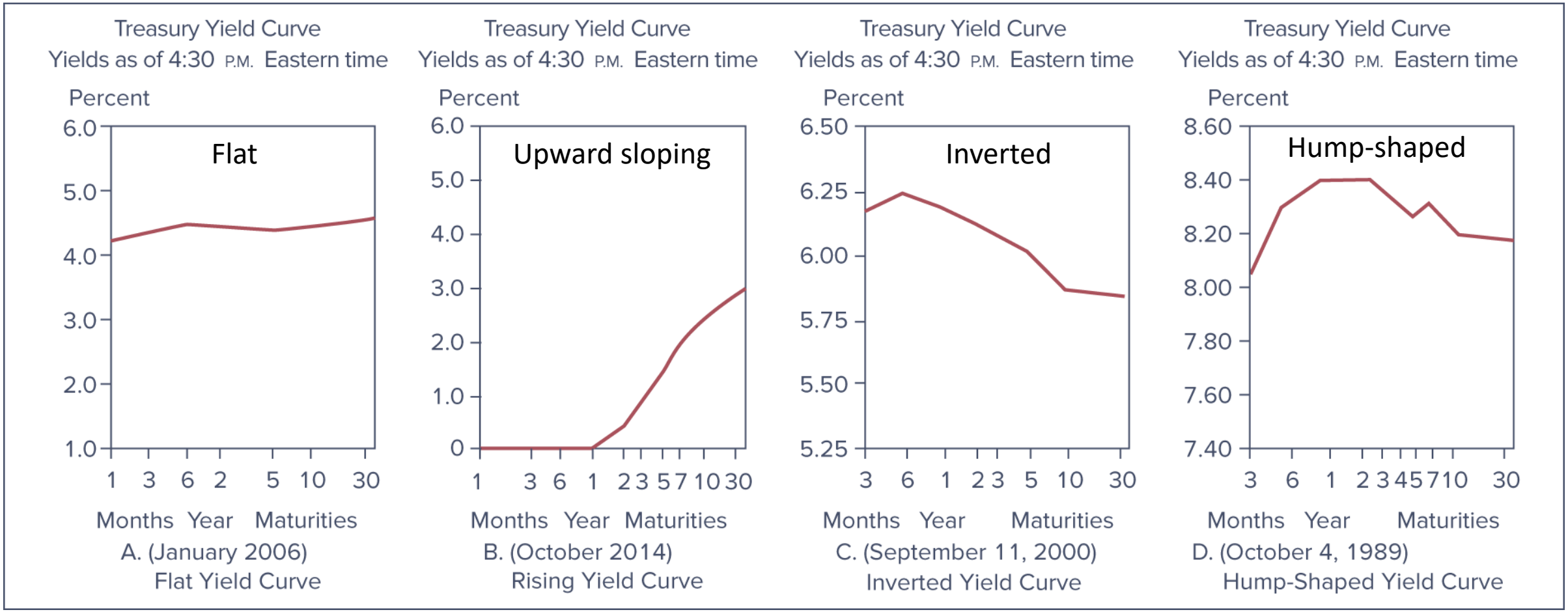
Should You Lock the Rate or Wait?

- You plan to invest money for 2 years and you have two choices:
  - **Option A: Lock now**
    - Take a 2-year rate today
  - **Option B: Wait and adjust**
    - Take a 1-year rate today
    - Then reinvest later
- Which strategy would you choose based on your expectations?

Term Structure of Interest Rates

- Term structure of interest rates:
  - Relationship between **yields to maturity** and **time to maturity** for otherwise comparable bonds.
- Yield curve:
  - The graphical representation of the term structure.

Common Yield Curve Shapes: Historical Examples



Expectations Theory

- Why might bonds with different maturities offer different yields?
  - Long-term yields reflect **expected future short-term interest rates**.
  - Therefore, expectations about future short-term rates help determine the **shape of the yield curve**.

Expectations Theory and the Yield Curve

- Expected future short-term rates **rise** → **Upward-sloping** yield curve
- Expected future short-term rates **fall** → **Downward-sloping / inverted** yield curve
- Expected future short-term rates **remain unchanged** → **Flat** yield curve

➢ If the market expects short-term rates to rise, why would today's longer-term yield already be higher?

How Does The Expectations Theory Work?

- Suppose you invest $1 today
  - Current 1-year interest rate = 8%.
  - Assume the market expects 1-year rate next year to be **10%**.
- Under Expectations Theory, **investing for 2 years today** and **rolling over two 1-year investments** should produce the same expected total return.
- What current 2-year rate makes these two strategies equivalent in expected return?

Two Investment Strategies Under Expectations Theory

A. **Lock in for 2 years:**

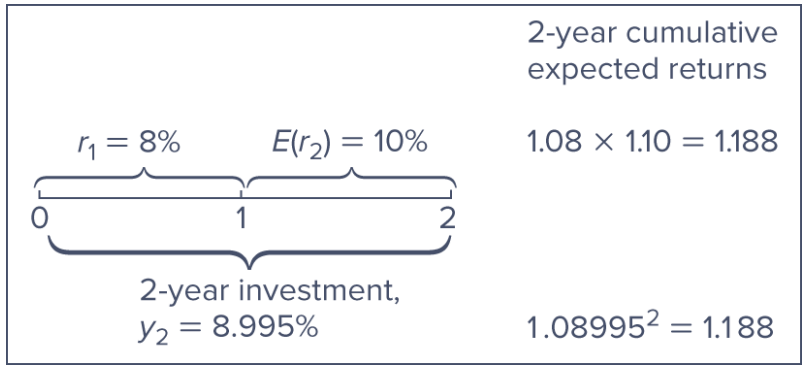
- Invest for 2 years at rate y_2.

$$\$1 \times (1 + y_2)^2$$

B. **Invest for 1 year, then reinvest:**

- Invest at 8% for 1 year, then reinvest at the expected 1-year rate next year of 10%:

$$\$1 \times (1 + 8\%) \times (1 + 10\%)$$



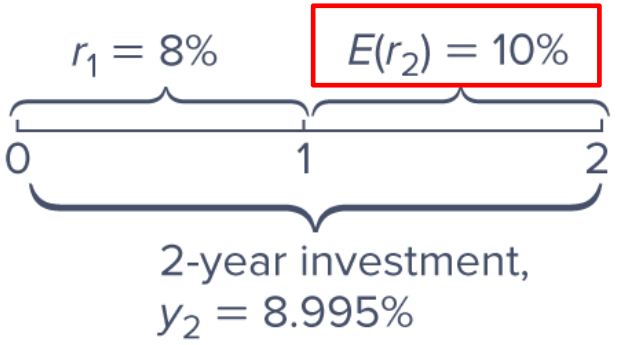
- **Under Expectations Theory:**

$$(1 + y_2)^2 = (1.08)(1.10)$$

$$y_2 = 8.995\% \approx 9.0\%$$

- Expected short-term rates rise → 2-year rate > today's 1-year rate.

Inferring Expected Future Rates



- We do not **directly observe** the market's expected future short-term rates.
- **Forward rate**: the rate for a future period implied by today's zero-coupon yields.
- Under Expectations Theory, the forward rate is **interpreted** as the market's expected future short-term rate.

Forward Rate Example

- Suppose a 2-year zero-coupon bond has a YTM of 6%
- A 3-year zero-coupon bond has a YTM of 7%.
- What is the implied 1-year forward rate for year 3?

Forward Rate Example

- **3-year investment at 7%:**

$$(1.07)^3$$

- **2-year investment at 6%**, then 1-year at f_3.

$$(1.06)^2(1+f_3)$$

- Set equal:

$$(1.07)^3 = (1.06)^2 \times (1+f_3)$$

$$f_3 = \frac{(1.07)^3}{(1.06)^2} - 1 = 9.028\%$$

- The implied 1-year forward rate for year 3 is 9.03%.
- Under Expectations Theory, this is interpreted as the expected 1-year rate in year 3.

Click here to explore the forward rate

One-Year Forward Rate: General Formula

$$(1+y_n)^n = (1+y_{n-1})^{n-1} \times (1+f_n)$$

- Thus, the implied one-year forward rate in year n is:

$$f_n = \frac{(1+y_n)^n}{(1+y_{n-1})^{n-1}} - 1$$

  - y_n = yield on an n-year zero-coupon bond
  - f_n = implied 1-year forward rate for year n

- Under Expectations Theory, f_n is interpreted as the expected 1-year rate in year n.

Practice: One-Year Forward Rate

- The current yield curve for default-free zero-coupon bonds is as follows:

Maturity (Years)	YTM
1	10%
2	11%
3	12%

- What is the implied 1-year forward rate for year 3, from the end of year 2 to the end of year 3?

Liquidity Preference Theory

- Longer-maturity bonds generally have greater price risk, so investors demand a premium.
- This additional compensation is called a **liquidity premium** or **term premium**.
- Therefore, longer-term yields may be higher even if expected future short-term rates do not rise.

Expectations vs. Liquidity Preference Theory

- **Expectations Theory:**
  Forward rate = expected future short-term rate
- Yield-curve slope reflects rate expectations.

- **Liquidity Preference Theory:**
  Forward rate = expected future short-term rate + liquidity premium
- Longer maturities may include a positive liquidity premium.

Implications for the Yield Curve

- **Upward-sloping curve**
  Higher expected future rates, a positive liquidity premium, or both.

- **Inverted curve**
  Stronger evidence that expected short-term rates are falling.

Expectations + Liquidity Premium:
Putting the Theories Together

- Under **Liquidity Preference** Theory, assume a **positive liquidity premium** that generally increases with maturity
- **Expected short-term rates rising**
  - ➢ Both forces push toward an **upward-sloping yield curve**
- **Expected short-term rates falling**
  - ➢ Liquidity premium offsets some or all of the downward pressure
  - ➢ Observed curve may be upward-sloping, flat, hump-shaped, or inverted.
- Yield-curve shape reflects both rate expectations and the liquidity premium.

Practice Questions for Exam 2

Chapter 10:

- 36. Under the expectations hypothesis…
- 37. The yield curve is upward-sloping…
- 39. Under the liquidity preference theory…
- 40. a. The yield curve for default-free zero-coupon bonds…

  b. Assume that the pure expectations hypothesis…
- 41. The yield to maturity on one-year zero-coupon bonds is 8%.
- 43. Consider the following $ 1,000 par value zero-coupon bonds.